

## Binary — Practice · AS 9618 — ANSWERS (student copy) · first edition, the sheet set as homework

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1. (a)  $91 = 01011011$  (b)  $172 = 10101100$  — 1 mark working, 1 mark answer each
2. (a) 179 (b) 126
3. (a) 256 (b) 65 536 (c) 16 777 216 ( $2^n$ )
4. 65 535 ( $= 2^{16} - 1$ )
5.  $154 + 102 = 256 = 1\ 00000000$  — needs 9 bits; the ninth bit is lost (overflow) [1]; register holds  $00000000 = 0$  [1]; result too large for 8 bits [1]
6.  $250 + 10 = 260$ ;  $260 - 256 = 4$  [1]; the addition overflows the 8-bit register, so the value wraps past zero [1]

 **The quiet one, unhinted.** Question 5 adds  $10011010 + 01100110 = 154 + 102 = \text{exactly } 256$  — so the eight-bit register does not hold a wrong number, it holds **00000000**. Nothing at all. The one addition where overflow leaves no trace whatsoever, which is far more alarming than question 6's  $250 + 10 = 4$ , and it is the same mechanism.